

Student Name: _____

Teacher: Ms Mandile / Mrs Underwood (circle)



Year 12 Extension 1

Wednesday 18th August 2021

Task 4 – Trial

General Instructions	
• Write using a black pen • Graphs may be drawn in pencil • No liquid paper or whiteout tape	
Reading Time	5 minutes
Working Time	120 minutes

Structure of the Paper		
Section I pages 2 – 5	10 marks	Allow about 15 minutes for this section
Section II pages 6 – 9	60 marks	Allow about 1 hour 45 minutes for this section
Total	70 marks	Weighting 30%

Special Instructions
• NESA approved Calculator may be used. • Attempt all questions • This examination must not be removed from the Examination Room. • Use the multiple choice answer sheet for section I • In Section II show relevant mathematical reasoning and/or calculations. • The mark allocation for each question provides a guide for the expected length of response. • Full marks may not be awarded for correct answers without working. • In section II answer each question in a separate writing booklet. Extra writing booklets are available if you need them. Number your answers clearly.

Section I

10 marks

Attempt Questions 1–10

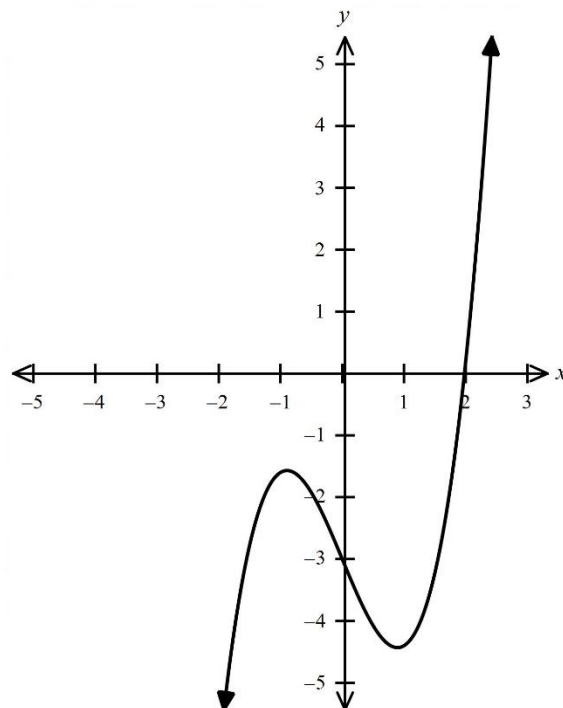
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1. Peter draws a vector from the origin to the point $A(-2, 5)$. He then draws a vector $-3\mathbf{i} + 2\mathbf{j}$ from A , ending at point B .

How far is point B from the origin?

- A. $\sqrt{10}$ units
B. $\sqrt{74}$ units
C. $\sqrt{13}$ units
D. $\sqrt{34}$ units
2. Consider the graph of $y = f(x)$ shown here:

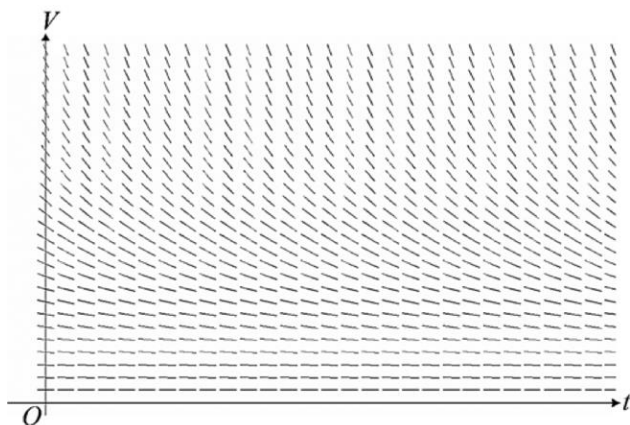


Which one of the following would have 2 more roots than $f(x)$?

- A. $y = -2 \times f(x)$
B. $y = f(x) + 3$
C. $y = f^{-1}(x)$
D. $y = f(x + 3)$

3. The volume of the solid formed when the area enclosed between the curve $y = x^3 - 1$, the y -axis, and the lines $y = 1$ and $y = 2$ is rotated about the y -axis is given by:
- A. $\pi \int_1^2 (x^3 - 1) dy$
- B. $\int_1^2 (\sqrt[3]{y+1}) dy$
- C. $\pi \int_1^2 (\sqrt[3]{y} + 1)^2 dy$
- D. $\pi \int_1^2 \sqrt[3]{(y+1)^2} dy$
4. Which one of the following differential equations is $y = 5xe^{2x}$ a solution?
- A. $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} = 0$
- B. $\frac{d^2y}{dx^2} - 4 \frac{dy}{dx} = 0$
- C. $\frac{d^2y}{dx^2} - 4y = 20e^{2x}$
- D. $\frac{d^2y}{dx^2} - 4y = e^{2x}$
5. The vectors $4\mathbf{i} - \mathbf{j}$ and $3\mathbf{i} + x\mathbf{j}$ are perpendicular. What is the value of x ?
- A. -12
- B. -1
- C. 12
- D. 13
6. Which of the following is an anti-derivative for $-\frac{1}{\sqrt{9-x^2}}$?
- A. $y = \tan^{-1} \left(\frac{x}{3} \right) + C$
- B. $y = \sin^{-1} 3x + C$
- C. $y = \cos^{-1} \left(\frac{x}{3} \right) + C$
- D. $y = \cos^{-1} 3x + C$

7. A direction field for the volume of water, V megalitres, in a reservoir t years after 2021 is shown below.



According to this model, for $k > 0$ what is the value of $\frac{dV}{dt}$?

- A. $-kV^2$
 - B. $-\frac{k}{V}$
 - C. $\frac{k}{V}$
 - D. kV^2
8. A company has a board of twelve directors. Six of these were selected at random to be candidates in the election for the positions of President, Vice-President and Treasurer.

In how many ways can these three senior positions be filled?

- A. ${}^{12}\mathbf{P}_3$
- B. $\frac{12!}{3!}$
- C. $\frac{{}^{12}\mathbf{P}_6}{3!}$
- D. $\frac{{}^{12}\mathbf{P}_6}{3!3!}$

9. A particle is initially positioned at $x = -2$. Its motion has velocity of $v = \frac{1}{t+e}$. Which of the following will be true when the particle reaches the origin?

A. $t = 2, v = \frac{1}{2+e}$

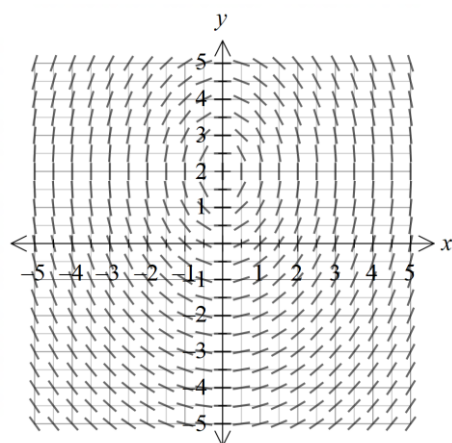
B. $t = e, \dot{x} = \frac{1}{2e}$

C. $t = e^2, v = \frac{1}{e^2+e}$

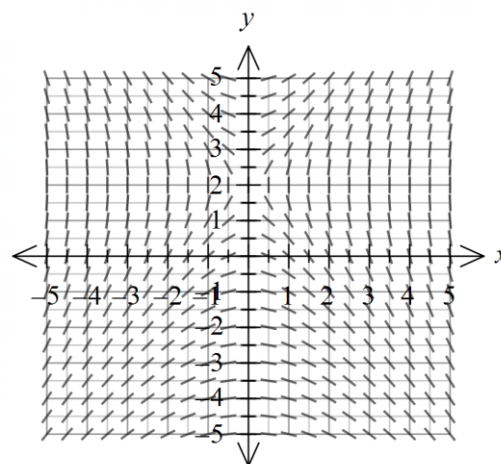
D. $t = e^3 - e, v = e^{-3}$

10. Which of the following best represents the direction field for the differential equation $\frac{dy}{dx} = \frac{-2x}{y-2}$?

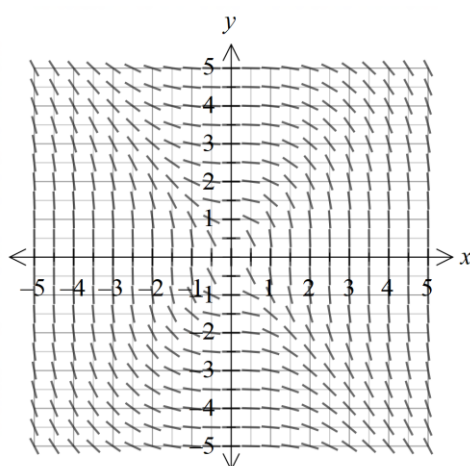
A.



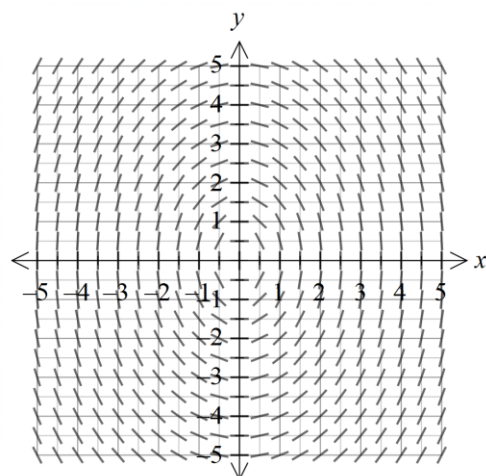
B.



C.



D.



Section II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new Writing Booklet

(a) Given $P(x) = x^3 + 11x^2 - 56x + 60$ and $P'(2) = 0$

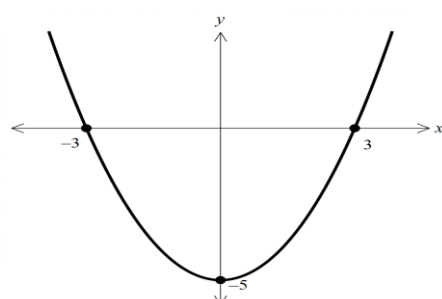
(i) Fully factorise the polynomial $P(x)$ 2

(ii) Hence, solve $P(x) \leq 0$ 1

(b) For what values of x is $\frac{x}{x-1} \geq -2$? 3

(c) Find the angle between the vectors $\mathbf{a} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$, correct to the nearest degree. 2

(d) The diagram shows the graph of $y = f(x)$.



(i) Sketch $y = f^{-1}(x)$ after making an appropriate restriction on the domain of $f(x)$. 1

(ii) Sketch $y = \frac{1}{f^{-1}(x)}$. 2

(e) By expressing $\sin x - \sqrt{3} \cos x$ in the form $A \sin(x - \alpha)$, solve $\sin x - \sqrt{3} \cos x = -\sqrt{2}$, for $0 \leq x \leq 2\pi$. 4

End of Question 11

Question 12 (15 marks) Start a new Writing Booklet

- (a) Use the principle of mathematical induction to show that for all integers $n \geq 1$, 3

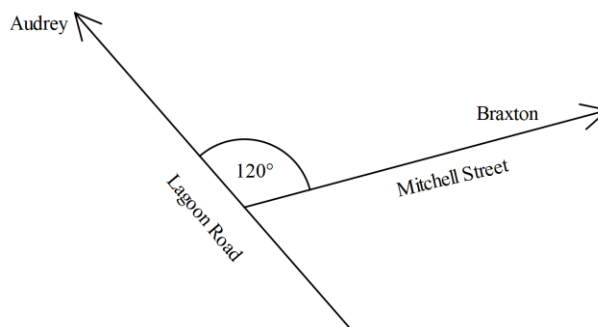
$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2}{4} (n+1)^2.$$

- (b) Use the substitution $x = u^2 - 1$, find the exact value of 3

$$\int_0^1 \frac{x}{\sqrt{x+1}} dx$$

- (c) Use the pigeonhole principle to determine how many integers must be selected from the numbers 5, 6, 7, 8, 9 and 10 so that at least 2 of the numbers will have a difference of 3? 2

- (d) Two straight roads, Lagoon Road and Mitchell Street, intersect at an angle of 120° .
Audrey leaves the intersection at 9:00 am and travels on Lagoon Road at a constant speed of 50 kilometres per hour.
Braxton leaves the intersection at 10:00 am, travelling along Mitchell Street at a constant speed of 60 kilometres per hour.
The diagram below shows the direction of Audrey and Braxton's travels.



Taking $t = 0$ at the time of Braxton's departure from the intersection:

- (i) Show that the distance between Audrey and Braxton at time, t hours can be expressed as $D = 10\sqrt{91t^2 + 80t + 25}$ kilometres. 2
- (ii) Find the rate of change in the distance between Audrey and Braxton at 12:00 noon, correct to the nearest km/h. 2
- (e) It is known that $\sin x = \frac{1}{3}$, where $\frac{\pi}{2} < x < \pi$. Find the exact value of $\cot 2x$? 3

End of Question 12

Question 13 (15 marks) Start a new Writing Booklet

- (a) Water is poured into a container at a rate of $8 \text{ cm}^3\text{s}^{-1}$ and the volume of the water in the container is given by: 3

$$V = \frac{3}{2}(h^2 + 8h) \quad \text{where } h \text{ cm is the depth of the water.}$$

Find the rate of change of the depth of the water when $V = 72 \text{ cm}^3$.

- (b) Using the substitution $u = x - 1$,
(i) show that: 1

$$\int \frac{x}{x^2 - 2x + 2} dx = \int \frac{u}{u^2 + 1} du + \int \frac{1}{u^2 + 1} du$$

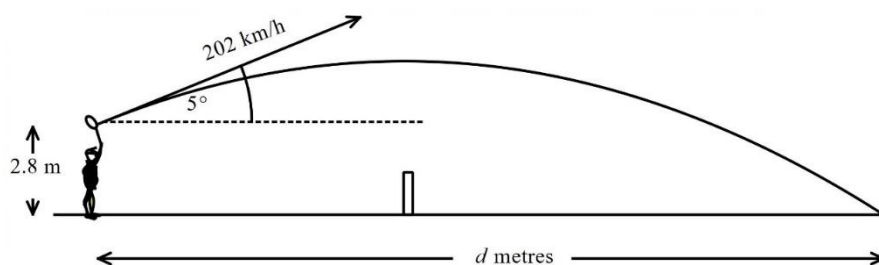
- (ii) Hence or otherwise find 2

$$\int \frac{x}{x^2 - 2x + 2} dx$$

- (c) Mick hits a tennis ball from a height 2.8 metres above the ground. The angle of elevation as the racket hits the ball is 5° and the velocity is 202 km/h.

202 km/h is approximately 56.1 m/s.

The ball lands at a horizontal distance of d metres from Mick's feet.



- (i) Using $g = 9.8 \text{ ms}^{-2}$ as acceleration due to gravity, show that the vector displacement of the tennis ball is: 3

$$\underline{s}(t) \approx 55.9t \underline{i} + (4.9t - 4.9t^2 + 2.8) \underline{j}.$$

- (ii) If d is greater than 26.2, the ball will be called “out”. 2

Find the value of d and determine if the ball will be called “out”.

Question 13 continues on page 9

Question 13 (continued)

- (d) A piece of hot metal is placed in a room with a surrounding air temperature of 15°C and allowed to cool. It loses heat according to Newton's law of cooling: 4

$$\frac{dT}{dt} = -k(T - A)$$

where T is the temperature of the metal in degrees Celsius at time t minutes, A is the surrounding air temperature and k is a positive constant.

After 5 minutes the temperature of the metal is 75°C , and after a further 3 minutes it is 45°C .

Solve the differential equation, expressing your answer in the form $T = f(t)$ and giving all constants in exact form.

End of Question 13**Question 14 (15 marks)** Start a new Writing Booklet

- (a) Consider the differential equation: $\frac{dy}{dx} = \frac{3x^2+4}{2y}$ 3

Find a solution to the equation in the form $y = f(x)$, given that $f(2) = 1$.

- (b) (i) Differentiate $x \cos^{-1} x$ with respect to x . 1

- (ii) Hence find $\int \cos^{-1} x \, dx$ 2

- (c) Given the function $f(x) = \sin^2 x + 2 \cos x$

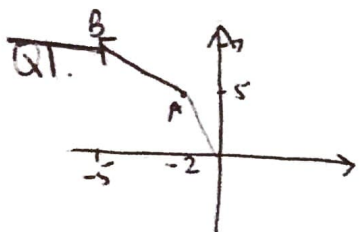
- (i) Determine if the function $y = f(x)$ is odd, even or neither. 2

- (ii) Show that the x -intercepts of $f(x) = \sin^2 x + 2 \cos x$ in the domain $-\pi \leq x \leq \pi$ are $x = \pm \cos^{-1}(1 - \sqrt{2})$ 3

- (iii) Hence or otherwise show that the exact area enclosed by the curve, $-\cos^{-1}(1 - \sqrt{2}) \leq x \leq \cos^{-1}(1 - \sqrt{2})$ and the x -axis is 4

$$\cos^{-1}(1 - \sqrt{2}) + (3 + \sqrt{2})\sqrt{2\sqrt{2} - 2}$$

End of Paper



$$OB = \sqrt{(5^2 + 7^2)} \\ = \sqrt{74}$$

B

B

Q2 $y = f(x) + 3$

Q3 $Vol = \pi \int x^2 dy$

$$y = x^3 - 1 \\ y + 1 = x^3$$

$$x = \sqrt[3]{(y+1)} \\ x^2 = (\sqrt[3]{(y+1)})^2$$

D

Q4 $y = 5x \cdot e^{2x}$

$$u = 5x \quad v = e^{2x} \\ u' = 5 \quad v' = 2e^{2x}$$

$$\frac{dy}{dx} = 5e^{2x} + 10xe^{2x}$$

$$\frac{d^2y}{dx^2} = 10e^{2x} + 2(5e^{2x} + 10xe^{2x}) \\ = 20e^{2x} + 20xe^{2x}$$

- not A or B

$$20e^{2x} + 20xe^{2x} - 4(5xe^{2x}) = 20e^{2x} \quad \therefore \quad C$$

Q5 If perpendicular then dot product = 0

$$4 \times 3 - x = 0$$

$$x = 12$$

C

C

Q6.

All gradients are negative since $K > 0$
then either A or B.

A

As V increases, $\frac{dV}{dt}$ gets steeper $\therefore$

$$\begin{aligned} \text{Q8. } {}^{12}C_6 \times {}^6P_3 &= \frac{12!}{6! 6!} \times \frac{6!}{3!} \\ &= \frac{{}^{12}P_6}{3!} \end{aligned}$$

C

$$\text{Q9. } \frac{dx}{dt} = \frac{1}{t+e}$$

$$x = \int \frac{1}{t+e} dt$$

$$x = \ln|t+e| + C$$

$$\text{When } t=0, x=-2$$

$$-2 = \ln e + C$$

$$C = -3$$

$$\therefore x = \ln(t+e) - 3$$

$$\text{When } x=0$$

$$\ln(t+e) = 3$$

$$t+e = e^3$$

$$t = e^3 - e$$

$\therefore D$

$$\text{Q10. } \frac{dy}{dx} = \frac{-2x}{y-2}$$

$$\text{at } (0,0) \quad \frac{dy}{dx} = 0 \quad \therefore \text{A or B}$$

$$\text{at } (-2,-2) \quad \frac{dy}{dx} = -\frac{2(-2)}{-2-2}$$

$$= -1$$

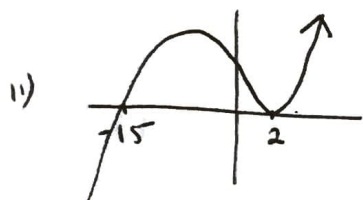
A

$$a) i) P(x) = x^3 + 11x^2 - 56x + 60$$

$$P'(2) = 0 \quad \therefore (x-2)^2 \text{ is a factor}$$

$$\begin{array}{r} x^2 - 4x + 4 \overline{) x^3 + 11x^2 - 56x + 60} \\ \underline{x^3 - 4x^2 + 4x} \\ 15x^2 - 60x + 60 \\ \underline{15x^2 - 60x + 60} \\ 0 \end{array}$$

$$\therefore P(x) = (x-2)^2(x+15)$$



$$P(x) \leq 0$$

$$\therefore x \leq -15, x = 2$$

$$b) \frac{x}{x-1} \geq -2 \quad x \neq 1$$

$$(x-1)^2 \times \frac{x}{x-1} \geq -2(x-1)^2$$

$$x(x-1) + 2(x-1)^2 \geq 0$$

$$(x-1)(x + 2x - 2) \geq 0$$

$$(x-1)(3x-2) \geq 0$$



$$\therefore x > 1, x \leq \frac{2}{3}$$

$$c) \vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$$

$$= x_1 x_1 + y_1 y_1$$

$$|\vec{a}| = \sqrt{9+25}$$

$$= \sqrt{34}$$

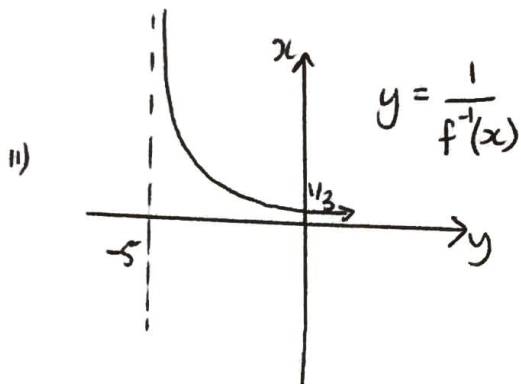
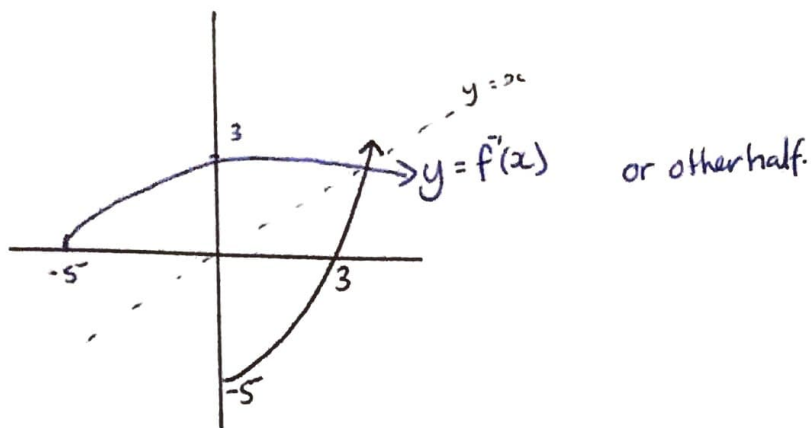
$$|\vec{b}| = \sqrt{1+16}$$

$$= \sqrt{17}$$

$$\therefore \sqrt{34} \cdot \sqrt{17} \cos \theta = -3 - 20$$

$$\cos \theta = -0.9566$$

$$\theta = 163^\circ$$



e)

$$\sin x - \sqrt{3} \cos x$$

$$A = \sqrt{1^2 + \sqrt{3}^2} = 2$$

$$\tan \alpha = \frac{\sqrt{3}}{1}$$

$$\alpha = \frac{\pi}{3}$$

$$\begin{cases} \cos \alpha = 1 \\ \sin \alpha = \sqrt{3} \end{cases}$$

$$\therefore \sin x - \sqrt{3} \cos x = 2 \sin \left(x - \frac{\pi}{3} \right)$$

$$2 \sin \left(x - \frac{\pi}{3} \right) = -\sqrt{2}$$

$$0 \leq x \leq 2\pi$$

$$\sin \left(x - \frac{\pi}{3} \right) = \frac{-\sqrt{2}}{2}$$

$$= -\frac{1}{\sqrt{2}}$$

$$-\frac{\pi}{3} \leq x - \frac{\pi}{3} \leq \frac{5\pi}{3}$$

$$\text{related acute } \left(x - \frac{\pi}{3} \right) = \frac{\pi}{4}$$



$$x - \frac{\pi}{3} = \pi + \frac{\pi}{4}, -\frac{\pi}{4}$$

$$x = \frac{\pi}{12}, \frac{19\pi}{12}$$

a) test $n=1$

$$\begin{aligned} \text{LHS} &= 1^3 \\ &= 1 \end{aligned} \quad \begin{aligned} \text{RHS} &= \frac{1}{4}(1+1)^2 \\ &= 1 \end{aligned}$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$S_k = \frac{k^2}{4}(k+1)^2$$

Prove true for $n=k+1$ i.e. show $S_{k+1} = \frac{(k+1)^2}{4}(k+2)^2$

Now $S_{k+1} = S_k + T_{k+1}$ using our assumption

$$\begin{aligned} &= \frac{k^2}{4}(k+1)^2 + (k+1)^3 \\ &= (k+1)^2 \left(\frac{k^2}{4} + k+1 \right) \\ &= \frac{(k+1)^2}{4} (k^2 + 4(k+1)) \\ &= \frac{(k+1)^2}{4} (k^2 + 4k + 4) \\ &= \frac{(k+1)^2}{4} (k+2)^2 \end{aligned}$$

$\therefore$ true for $n=k+1$

By the principle of mathematical induction the above statement is proved to be true.

$$x = u^2 - 1$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

$$\begin{aligned} \text{When } x &= 0 \\ u^2 - 1 &= 0 \\ u &= 1 \end{aligned}$$

$$\begin{aligned} x &= 1 \\ u^2 - 1 &= 1 \\ u &= \sqrt{2} \end{aligned}$$

$$\int_0^1 \frac{x}{\sqrt{x+1}} dx = \int_1^{\sqrt{2}} \frac{u^2 - 1}{\sqrt{u^2 - 1 + 1}} 2u du$$

$$= \int_1^{\sqrt{2}} \frac{u^2 - 1}{u} \cdot 2u du$$

$$= 2 \int_1^{\sqrt{2}} (u^2 - 1) du$$

$$= 2 \left[\frac{u^3}{3} - u \right]_1^{\sqrt{2}}$$

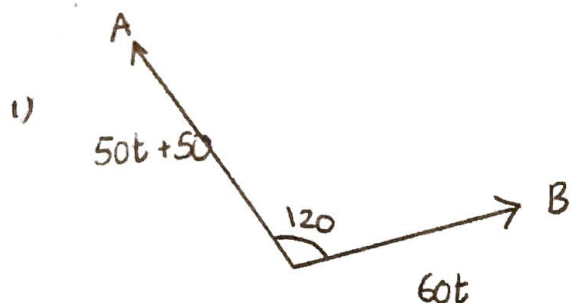
$$= 2 \left\{ \left(\frac{2\sqrt{2}}{3} - \sqrt{2} \right) - \left(\frac{1}{3} - 1 \right) \right\}$$

$$= 2 \left\{ \frac{2\sqrt{2}}{3} - \frac{3\sqrt{2}}{3} + \frac{2}{3} \right\}$$

$$= \frac{2(2 - \sqrt{2})}{3}$$

c) If we select 5, 6, 7 then there is no difference of 3

$\therefore$ We need 4 digits to be selected.



$$D^2 = (50(t+1))^2 + (60t)^2 - 2 \times 50(t+1) \times 60t \cos 120$$

$$= 2500(t^2 + 2t + 1) + 3600t^2 + 3000t(t+1)$$

$$= 2500t^2 + 5000t + 2500 + 3600t^2 + 3000t^2 + 3000t$$

$$= 9100t^2 + 8000t + 2500$$

$$= 100(91t^2 + 80t + 25)$$

$$D = \sqrt{100(91t^2 + 80t + 25)}$$

$$= 10 \sqrt{91t^2 + 80t + 25}$$

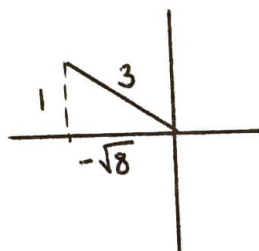
$$11) \frac{dD}{dt} = 10 \times \frac{1}{2} (182t + 80) (91t^2 + 80t + 25)^{-1/2}$$

When $t = 2$

$$\frac{dD}{dt} = 94.747$$

$$= 95 \text{ km/h}$$

$$e) \sin x = \frac{1}{3} \quad \frac{\pi}{2} < x < \pi$$



$$\tan x = -\frac{1}{\sqrt{8}}$$

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\cot 2x = \frac{1 - \tan^2 x}{2 \tan x} = \frac{1 - \left(-\frac{1}{\sqrt{8}}\right)^2}{2 \times \frac{1}{\sqrt{8}}} = \frac{-7\sqrt{8}}{16} = \frac{-7 \times 2\sqrt{2}}{16 \times 8} = -\frac{7\sqrt{2}}{8}$$

$$a) \quad \frac{dv}{dt} = 8 \text{ cm}^3 \text{ s}^{-1}$$

$$V = \frac{3}{2}(h^2 + 8h)$$

$$\frac{dh}{dt} = \frac{dv}{dt} \cdot \frac{dh}{dV}$$

$$\frac{dv}{dh} = \frac{3}{2}(2h + 8) \\ = 3h + 12$$

$$\frac{dh}{dV} = \frac{1}{3h + 12} \quad \checkmark$$

= .

When $V = 72$

$$\frac{3}{2}(h^2 + 8h) = 72$$

$$h^2 + 8h - 48 = 0$$

$$(h + 12)(h - 4) = 0$$

$$h = -12, 4 \quad \checkmark$$

↑
invalid
solⁿ

$$\therefore \frac{dh}{dt} = 8 \times \frac{1}{3 \times 4 + 12} \quad \checkmark$$

$$= \frac{1}{3} \text{ cm s}^{-1}$$

(3)

$$b) \quad i) \quad u = x - 1$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$\int \frac{x}{x^2 - 2x + 2} dx = \int \frac{u + 1}{u^2 + 1} du$$

$$= \int \frac{u}{u^2 + 1} du + \int \frac{1}{u^2 + 1} du$$

(1)

$$ii) \quad \int \frac{x}{x^2 - 2x + 2} dx = \frac{1}{2} \int \frac{2u}{u^2 + 1} du + \int \frac{1}{u^2 + 1} du$$

$$= \frac{1}{2} \ln|u^2 + 1| + \tan^{-1}(u) + C \quad \checkmark$$

$$= \frac{1}{2} \ln|(x-1)^2 + 1| + \tan^{-1}(x-1) + C$$

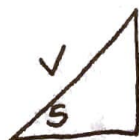
$$= \frac{1}{2} \ln|x^2 - 2x + 2| + \tan^{-1}(x-1) + C \quad \checkmark$$

(2)

$$V = 202 \text{ Km/h}$$

$$= \frac{202 \times 1000}{3600}$$

$$= 56.119 \text{ m s}^{-1}$$



$$\dot{y} = 56.1 \sin 5$$

$$= 4.9$$

$$\dot{x} = 56.1 \cos 5$$

$$= 55.9$$

$$\underline{a}(t) = \ddot{x} \underline{i} + \ddot{y} \underline{j}$$

$$= 0 \underline{i} - 9.8 \underline{j}$$

$$\underline{v}(t) = \int 0 \underline{i} - 9.8 \underline{j} \, dt$$

$$= C \underline{i} + (-9.8t + C_1) \underline{j}$$

When $t=0$

$$C = 55.9$$

$$C_1 = 4.9$$

$$\underline{v}(t) = 55.9 \underline{i} + (-9.8t + 4.9) \underline{j}$$

$$\underline{s}(t) = \int 55.9 \underline{i} + (-9.8t + 4.9) \underline{j} \, dt$$

$$= (55.9t + C_2) \underline{i} + (-4.9t^2 + 4.9t + C_3) \underline{j}$$

When $t=0$

$$x=0 \quad \therefore C_2 = 0$$

$$y = 2.8$$

$$C_3 = 2.8$$

$$\underline{s}(t) = 55.9t \underline{i} + (-4.9t^2 + 4.9t + 2.8) \underline{j}$$

ii) Ball hits the ground when vertical displacement is zero

$$-4.9t^2 + 4.9t + 2.8 = 0$$

$$t = \frac{-4.9 \pm \sqrt{(4.9)^2 - 4(-4.9)(2.8)}}{2 \times -4.9}$$

$$= -0.4, 1.41 \quad t > 0 \therefore t = 1.41$$

$$\therefore d = 55.9 \times 1.41$$

$$= 78.6 \text{ m}$$

$\therefore$ ball is out

(3)

(2)

$$\frac{dT}{dt} = -K(T-15)$$

When $t=5$ $T=75$

$t=8$ $T=45$

$$\frac{dt}{dT} = -\frac{1}{K(T-15)}$$

$$t = -\frac{1}{K} \int \frac{1}{T-15} dT$$

$$-Kt = \ln|T-15| + C \quad \checkmark$$

$$= \ln|T-15| + \ln e^C$$

$$= \ln e^C(T-15)$$

$$\therefore e^{-Kt} = e^C(T-15)$$

$$= B(T-15) \quad \text{where } B = e^C$$

$$\therefore e^{-5K} = 60B \quad \dots (1)$$

$$e^{-8K} = 30B \quad \dots (2)$$

$$(1) \div (2)$$

$$e^{3K} = 2$$

$$3K = \ln 2$$

$$K = \frac{\ln 2}{3} \quad \checkmark$$

$$\therefore \frac{1}{B} e^{-t \cdot \frac{\ln 2}{3}} = T-15$$

$$T = 15 + \frac{1}{B} e^{-t \cdot \frac{\ln 2}{3}}$$

continued

$$B = \frac{e^{-\frac{\ln 2}{3} \cdot t}}{T-15}$$

when $t=5$ $T=75$

$$B = \frac{e^{-\frac{5 \ln 2}{3}}}{60}$$

$$= \frac{e^{\ln 2^{-5/3}}}{60}$$

$$= \frac{2^{-5/3}}{60}$$

$$= \frac{1}{120\sqrt[3]{4}} \quad \checkmark$$

$$\therefore T = 15 + 120\sqrt[3]{4} e^{-t \cdot \frac{\ln 2}{3}} \quad \checkmark$$

Q14.

a) $\frac{dy}{dx} = \frac{3x^2+4}{2y}$

$$\int 2y dy = \int (3x^2+4) dx \quad \checkmark$$

$$y^2 = x^3 + 4x + C$$

$$f(2) = 1$$

$$1^2 = 2^3 + 4 \times 2 + C$$

$$C = -15 \quad \checkmark$$

$$\therefore y^2 = x^3 + 4x - 15 \quad \checkmark$$

$$y = \pm \sqrt{x^3 + 4x - 15}$$

but $f(2)=1 \therefore y = \sqrt{x^3 + 4x - 15}$

$$f(2)=1$$

$$\frac{d}{dx}(x \cos^{-1} x)$$

$$u = x \quad v = \cos^{-1} x$$

$$u' = 1 \quad v' = -\frac{1}{\sqrt{1-x^2}}$$

$$= 1 \times \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

$$= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

(1)

$$ii) \therefore \int \frac{d}{dx}(x \cos^{-1} x) dx = \int \cos^{-1} x dx + \int \frac{-x}{\sqrt{1-x^2}} dx$$

$$x \cos^{-1} x = \int \cos^{-1} x dx + \frac{1}{2} \int -2x(1-x^2)^{-1/2} dx \quad \checkmark$$

$$\int \cos^{-1} x dx = x \cos^{-1} x - \frac{1}{2} \frac{(1-x^2)^{1/2}}{1/2} + C$$

$$= x \cos^{-1} x - \sqrt{1-x^2} + C \quad \checkmark$$

(2)

$$c) i) f(x) = \sin^2 x + 2 \cos x$$

$$f(-x) = \sin^2(-x) + 2 \cos(-x)$$

$$= \sin^2 x + 2 \cos x \quad \checkmark$$

(2)

$\therefore f(x) = f(-x)$ so the function is even.

$$ii) \sin^2 x + 2 \cos x = 0$$

$$1 - \cos^2 x + 2 \cos x = 0$$

$$\cos^2 x - 2 \cos x - 1 = 0$$

$$\cos x = \frac{2 \pm \sqrt{2^2 - 4 \times 1 \times (-1)}}{2}$$

continued

$$\cos x = 1 \pm \sqrt{2} \quad \checkmark$$

but $\cos x \neq 1 + \sqrt{2}$

$$\therefore \cos x = 1 - \sqrt{2}$$

$$x = \cos^{-1}(1 - \sqrt{2}) \quad \checkmark$$

but since $f(x)$ is even, it is symmetrical about y-axis

$$\therefore x \text{ intercepts are } x = \pm \cos^{-1}(1 - \sqrt{2}) \quad \checkmark$$

(3)

$$(iii) \text{ Area} = \int_{-\cos^{-1}(1-\sqrt{2})}^{\cos^{-1}(1+\sqrt{2})} (\sin^2 x + 2 \cos x) dx$$

$$= 2 \int_0^{\cos^{-1}(1+\sqrt{2})} \sin^2 x + 2 \cos x dx \quad \checkmark$$

$$= 2 \int_0^{\cos^{-1}(1+\sqrt{2})} \left(\frac{1}{2} (1 - \cos 2x) + 2 \cos x \right) dx$$

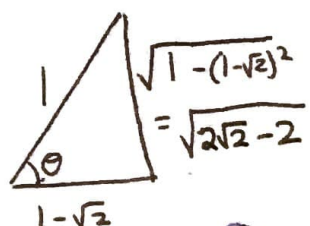
$$= 2 \left[\frac{1}{2} (x - \frac{1}{2} \sin 2x) + 2 \sin x \right]_0^{\cos^{-1}(1-\sqrt{2})}$$

$$= \left[x - \frac{1}{2} \sin 2x + 4 \sin x \right]_0^{\cos^{-1}(1-\sqrt{2})} \quad \checkmark$$

$$= \left[x - \sin x \cos x + 4 \sin x \right]_0^{\cos^{-1}(1-\sqrt{2})}$$

$$= \cos^{-1}(1-\sqrt{2}) - \sin(\cos^{-1}(1-\sqrt{2})) \cdot \cos(\cos^{-1}(1-\sqrt{2})) + 4 \sin(\cos^{-1}(1-\sqrt{2})) \quad \checkmark$$

$$\begin{aligned} &= \cos^{-1}(1-\sqrt{2}) - \sqrt{2\sqrt{2}-2} \cdot (1-\sqrt{2}) + 4\sqrt{2\sqrt{2}-2} \\ &= \cos^{-1}(1-\sqrt{2}) + \sqrt{2\sqrt{2}-2} (\sqrt{2}-1+4) \\ &= \cos^{-1}(1-\sqrt{2}) + \sqrt{2\sqrt{2}-2} (3+\sqrt{2}) \quad \checkmark \end{aligned}$$



(4)